

ISHAAN KATARA

50 Kiran Vihar, New Delhi

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Summary

Software Engineer with experience in **building** and **deploying scalable ML applications and backend systems**. Proficient in **Software Development**, **Agentic AI**, and **DevSecOps**, with a focus on delivering high performance software solutions.

Education

Jaypee Institute of Information Technology, India

Sep. 2021 – July 2025

Bachelor of Technology (B.Tech) in Computer Science Engineering

Noida, Uttar Pradesh

Technical Skills

Languages: Python, JavaScript, Typescript, C, C++, Rust, SQL

Frameworks: PyTorch, LangChain, Node.js/NestJS, React.js/NextJS, Tauri, FastAPI, Flask, ONNX

DevSecOps: AWS, Docker, Kubernetes, Jenkins, GitLab CI, GitHub Actions, Ansible, SonarQube, Terraform

Databases & Version Control: MongoDB, PostgreSQL, SQLite, Git

Soft Skills: Problem Solving, Communication, Leadership, Critical Thinking, Adaptability, Collaboration, Growth Mindset

Experience

Ministry of Electronics and Information Technology

May 2024 – August 2024

Machine Learning SWE Intern

New Delhi, Delhi

- Engineered an **NLP-powered** chatbot using **PyTorch** to automate inquiry categorization.
- Actively engaged with team members to understand inquiry categorization requirements.
- Collaborated with cross-functional teams and achieved a **47% increase** in successfully handled user queries.
- Implemented a complete **MLOps** pipeline using **GitHub Actions** for CI/CD, automating the build and testing.
- Deployed the application to **AWS EC2**, ensuring rapid, consistent updates.
- Technologies** Python, PyTorch, NLTK, RestAPI, HTML5, JavaScript, Tailwind CSS, Flask, AWS, GitHub Actions.

Sustainivo

March 2024 – May 2024

SDE Back-End Intern

Kolkata, West Bengal

- Collaborated with the backend team and implemented a secure **token-based authentication system** using **JWT**.
- Safeguarded user data and secured critical **API endpoints** across 3 core **microservices**.
- Established a scalable and high-performance production environment on a **virtual Ubuntu server**.
- Utilized **Nginx** as a **reverse proxy** and **Gunicorn** as the **WSGI server** for the **Flask** application.
- Technologies:** Python, Node.js, Gunicorn, Nginx, Shell Scripting, Flask, AWS, GitHub Actions.

Projects

R.I.M.A.S (Real-Time Intelligent Monitoring Advanced System)

[GitHub](#)

A cross-platform **Agentic AI** that integrates **Computer Vision**, **ML**, and voice recognition to help users improve productivity.

- Developed a multi-modal perception system by **fine-tuning** the **YOLOv8** model for real-time distraction detection.
- Fine-tuned LLaMA-2-7b** model for generating AI-driven productivity feedback.
- Architected a high-performance, cross-platform desktop application using **Tauri** and **Rust** as the UI for ML model.
- Tech Stack:** Tauri, Rust, YOLOv8, OpenCV, Python, FastAPI, PyTorch, LLaMA-2-7b, SQLite, TTS libraries, MLOps

XATNYS: AI-Powered Operating System

[GitHub](#)

A **microkernel architecture** operating system having full integration of **AI** at the frontend and backend levels of the system.

- Architected a **microkernel-based OS** from scratch in **Rust**.
- In a controlled testing environment, this design demonstrated up to a **28% faster application launch time** and **23% improvement in power efficiency** compared to a simple baseline model.
- Tech Stack:** Rust, Python, In-Line Assembly, PyTorch, CNN, Reinforcement Learning, NLP, ONNX, Docker

HashKrypt: Zero-Knowledge File Encryption Platform - [Link](#)

[GitHub](#)

A **secure** web application enabling file encryption, sharing, and decryption with **zero-knowledge key management**.

- Designed client-side **AES-256 encryption** with personal key-based vault unlocking and **asymmetric** key wrapping for secure sharing.
- Implemented **zero-knowledge** model ensuring server never accesses plaintext keys or files; integrated **Railway (Postgres)** for server hosting and **Vercel** for client-side deployment.
- Tech Stack:** Typescript, Next.js, NestJS, GitHub Actions, Prisma, PostgreSQL, Railway, Vercel, Redis, Tailwind CSS, BullMQ, WebCrypto/libsodium